

FFT-Based Fraunhofer Diffraction

Algorithms, Sampling Design, and Error Analysis

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Main Question

Fraunhofer diffraction is formally a Fourier transform, so the FFT is the natural numerical tool. The central issue is that the FFT also imposes a sampling grid.

Goal

Build a reliable FFT diffraction solver by connecting physical approximations, grid design rules, error analysis, and numerical experiments.

The final physical target is to recover the Rayleigh resolution scale $1.22\lambda/D$ from the computed Airy pattern.

- ① Physical model: Helmholtz $\rightarrow$ angular spectrum $\rightarrow$ Fraunhofer.
- ② Numerical discretization: continuous Fourier integral $\rightarrow$ DFT $\rightarrow$ screen grid.
- ③ Algorithms: FFT propagator, Simpson reference, ASM reference, Rayleigh recovery.
- ④ Error analysis: E1 aliasing, E2 screen resolution, E3 model error.
- ⑤ Experiments: analytical validation, convergence, grid rules, breakdown, efficiency, Rayleigh.

Fouier Transform Convention

We use the transform pair

$$\hat{f}(\xi, \eta) = \int_{\mathbb{R}^2} f(x, y) e^{-2\pi i(x\xi + y\eta)} dx dy,$$

$$f(x, y) = \int_{\mathbb{R}^2} \hat{f}(\xi, \eta) e^{2\pi i(x\xi + y\eta)} d\xi d\eta.$$

With this convention, partial derivatives become multiplication in the frequency domain:

$$\widehat{\frac{\partial f}{\partial x}}(\xi, \eta) = 2\pi i \xi \hat{f}(\xi, \eta), \quad \widehat{\frac{\partial f}{\partial y}}(\xi, \eta) = 2\pi i \eta \hat{f}(\xi, \eta).$$

Hence

$$\widehat{\frac{\partial^2 f}{\partial x^2}}(\xi, \eta) = -4\pi^2 \xi^2 \hat{f}(\xi, \eta), \quad \widehat{\frac{\partial^2 f}{\partial y^2}}(\xi, \eta) = -4\pi^2 \eta^2 \hat{f}(\xi, \eta).$$

Scalar Wave Model

A monochromatic scalar field is written as

$$U(x, y, z, t) = U(x, y, z)e^{-i\omega t}.$$

In a homogeneous source-free region, the spatial part satisfies the Helmholtz equation

$$\Delta U + k^2 U = 0, \quad k = \frac{2\pi}{\lambda}.$$

The aperture field $A(x, y) = U(x, y, 0)$ is propagated from the aperture plane to a screen at distance z .

Angular Spectrum Representation

Taking the two-dimensional Fourier transform in x, y diagonalizes the transverse derivatives. For each spatial frequency (f_x, f_y) , the Helmholtz equation becomes

$$\frac{\partial^2 \hat{U}}{\partial z^2} + k_z(f_x, f_y)^2 \hat{U} = 0, \quad k_z^2 = 4\pi^2 \left(\frac{1}{\lambda^2} - f_x^2 - f_y^2 \right).$$

The transfer function is therefore

$$H(f_x, f_y, z) = \begin{cases} e^{ik_z(f_x, f_y)z}, & f_x^2 + f_y^2 < \lambda^{-2}, \\ e^{-\sqrt{-k_z(f_x, f_y)^2}z}, & f_x^2 + f_y^2 > \lambda^{-2}. \end{cases}$$

Hence the angular-spectrum solution is $U(x, y, z) = \mathcal{F}^{-1}\{\hat{A}(f_x, f_y)H(f_x, f_y, z)\}$.

Approximation Ladder

The physical models form a hierarchy:

Model	Valid regime	Numerical role
Angular spectrum	general propagation	reference
Fresnel	paraxial propagation	near/far transition
Fraunhofer	far field, small Fresnel number	main FFT solver

The Fresnel number $F = a^2/(\lambda z)$ is the key dimensionless quantity controlling the Fraunhofer model error.

Fresnel Approximation

For paraxial waves, $f_x^2 + f_y^2 \ll \lambda^{-2}$, so

$$\sqrt{\lambda^{-2} - (f_x^2 + f_y^2)} \approx \frac{1}{\lambda} - \frac{\lambda}{2}(f_x^2 + f_y^2).$$

This gives the Fresnel transfer function

$$H_F(f_x, f_y; z) = e^{ikz} e^{-i\pi\lambda z(f_x^2 + f_y^2)}.$$

Fraunhofer is obtained from Fresnel by further neglecting the aperture-plane quadratic phase in the far-field regime.

Fraunhofer Formula

In the far field, the screen field is proportional to the Fourier transform of the aperture:

$$U(X, Y; z) \propto \hat{A}\left(\frac{X}{\lambda z}, \frac{Y}{\lambda z}\right).$$

Thus the frequency-screen correspondence is

$$X = \lambda z f_x, \quad Y = \lambda z f_y.$$

This simple relation is what converts the FFT frequency grid into a physical screen grid.

From the Aperture Grid to FFT Frequencies

Use a square computational window $W_L = [-L/2, L/2)^2$, with $L = N\Delta x$. The centered aperture grid is

$$x_m = \left(m - \frac{N}{2}\right) \Delta x, \quad y_n = \left(n - \frac{N}{2}\right) \Delta x, \quad 0 \leq m, n \leq N-1.$$

The Fourier transform is approximated by the rectangle rule:

$$\hat{A}(f_x, f_y) \approx \Delta x^2 \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} A_{m,n} e^{-2\pi i(x_m f_x + y_n f_y)}.$$

Matching this phase with the DFT kernel forces $f_x \Delta x = p/N$ and $f_y \Delta x = q/N$. In centered order,

$$f_p = \frac{p - N/2}{N\Delta x}, \quad f_q = \frac{q - N/2}{N\Delta x}.$$

Centered FFT Convention

The centered physical grid introduces a phase convention, handled in code by shifts:

$$\hat{A}(f_p, f_q) \approx \Delta x^2 \text{fftshift}(\text{fft2}(\text{ifftshift}(A)))_{p,q}.$$

- `ifftshift`: move the physical center to the DFT origin before applying FFT.
- `fftshift`: put zero frequency back at the center for plotting.
- The FFT is still the rectangle-rule approximation of the continuous Fourier transform.

Screen Coordinates Induced by the FFT

Fraunhofer requires $X/(\lambda z) = f_x$, so

$$X_p = \lambda z f_p, \quad Y_q = \lambda z f_q.$$

The three key grid relations are

$$\Delta X = \frac{\lambda z}{L}, \quad X_{\max} \approx \frac{\lambda z}{2\Delta x}, \quad N = \frac{L}{\Delta x}.$$

Interpretation

L controls screen resolution; Δx controls observable screen range.

Grid Design Rules

For a target screen range $X_{\max}^{\text{tgt}}$, target screen spacing ΔX^{tgt} , and aperture scale a , use

$$\Delta x \leq \frac{\lambda z}{2X_{\max}^{\text{tgt}}}, \quad L \geq \frac{\lambda z}{\Delta X^{\text{tgt}}}.$$

$$L \geq 2a, \quad \Delta x \leq \frac{a}{k_{\text{edge}}}.$$

A practical choice is

$$\Delta x = \min \left\{ \frac{\lambda z}{2X_{\max}^{\text{tgt}}}, \frac{a}{k_{\text{edge}}} \right\}, \quad L_{\min} = \max \left\{ 2a, \frac{\lambda z}{\Delta X^{\text{tgt}}} \right\}.$$

Then choose N so that $N\Delta x \geq L_{\min}$, usually rounded to a convenient FFT size.

What Increasing N Really Means

Fixed Δx

$L = N\Delta x$ increases.

$$\Delta X = \frac{\lambda z}{L} \downarrow, \quad X_{\max} \text{ unchanged.}$$

Fixed L

$\Delta x = L/N$ decreases.

This refines aperture sampling:

$$X_{\max} \approx \frac{\lambda z}{2\Delta x} \uparrow, \quad \Delta X \text{ unchanged.}$$

Thus aliasing is controlled mainly by Δx , while screen resolution is controlled mainly by L .

Algorithm Roles

Method	Cost	Purpose
Fraunhofer FFT	$O(N^2 \log N)$	main far-field solver
Simpson quadrature	$O(N^4)$	direct integral check
Angular spectrum	$O(N^2 \log N)$	propagation reference
Rayleigh recovery	post-processing	physical validation

The reference methods prevent the project from relying on a single implementation of the FFT formula.

Fraunhofer FFT Propagator

Given a centered aperture array A , the main computation is

$$U_{p,q}^{\text{Fh}} = \Delta x^2 \text{fftshift}(\text{fft2}(\text{ifftshift}(A)))_{p,q}.$$

The screen coordinates are

$$X_p = \lambda z \frac{p - N/2}{N \Delta x}, \quad Y_q = \lambda z \frac{q - N/2}{N \Delta x}.$$

Intensities are compared after peak normalization, so the global factor $1/(\lambda^2 z^2)$ is omitted in most experiments.

Reference Computations and Rayleigh Recovery

- **Simpson quadrature:** evaluates the Fraunhofer integral directly at prescribed screen points. On an $N \times N$ aperture and an $N \times N$ screen, the cost is $O(N^4)$.
- **ASM:** computes $U(\cdot, \cdot, z) = \mathcal{F}^{-1}\{\mathcal{F}\{A\}H\}$, so it is used as a propagation reference when Fraunhofer may fail.
- **Rayleigh recovery:** compute the Airy pattern, extract the central cut, locate the first dark ring, and return

$$c_{\text{num}} = \frac{r_1 D}{\lambda z}.$$

The expected circular-aperture value is $c = 1.2196699 \dots$

Intensities are peak-normalized before comparing shapes:

$$\bar{I} = \frac{I}{\max I}.$$

The main relative error is

$$\text{Err}_2(I_{\text{num}}, I_{\text{ref}}) = \frac{\|\bar{I}_{\text{num}} - \bar{I}_{\text{ref}}\|_2}{\|\bar{I}_{\text{ref}}\|_2}.$$

We also compare shape features such as FWHM, first-null location, and the Rayleigh constant when those are more physically meaningful.

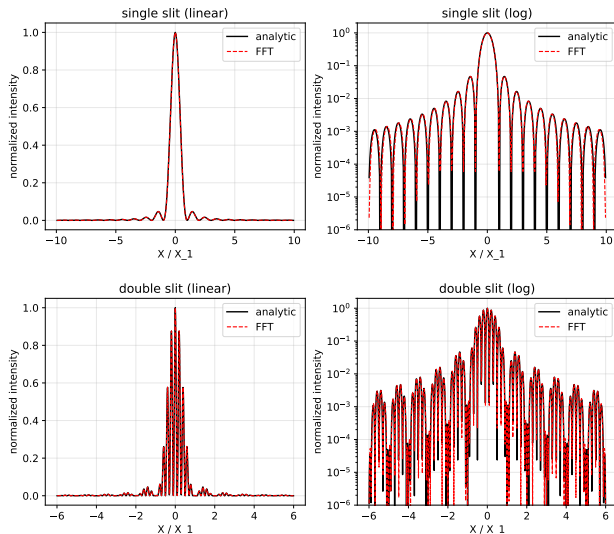
Error Decomposition

Symbol	Meaning	Controlled by
E1	aliasing error	Δx , through $X_{\max} \approx \lambda z / (2\Delta x)$
E2	screen-resolution error	L , through $\Delta X = \lambda z / L$
E3	Fraunhofer model error	$F = a^2 / (\lambda z)$

- E1 folds unresolved high-frequency content into the computed pattern.
- E2 under-samples fringes, nulls, or sidelobes on the screen.
- E3 comes from dropping the Fresnel quadratic phase; its largest omitted phase is $\Delta\phi_{\max} = \pi F$.

Exp.	Purpose	Main mechanism
1	analytical validation	normalization and coordinates
2	convergence rates	smoothness of aperture
3	grid-design tightness	E1 and E2
4	Fraunhofer breakdown	E3 and Fresnel number
5	efficiency comparison	FFT versus quadrature
6	Rayleigh recovery	physical resolution limit
7	complex apertures	qualitative Fourier optics

Experiment 1: Analytical Validation

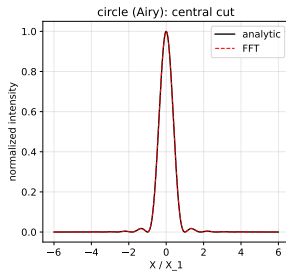
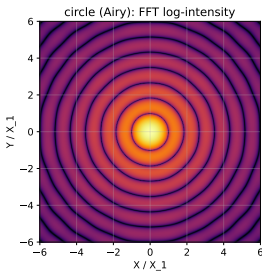
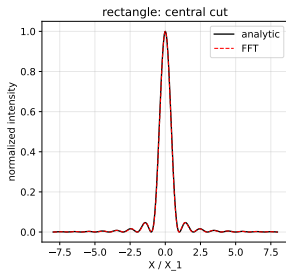
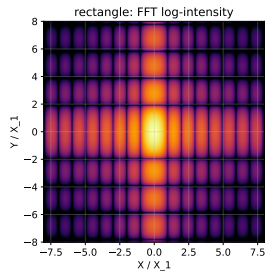


Closed-form tests:

- single slit: sinc^2
- double slit: $\text{sinc}^2 \cos^2$
- rectangle: product of sinc terms
- circle: Airy pattern

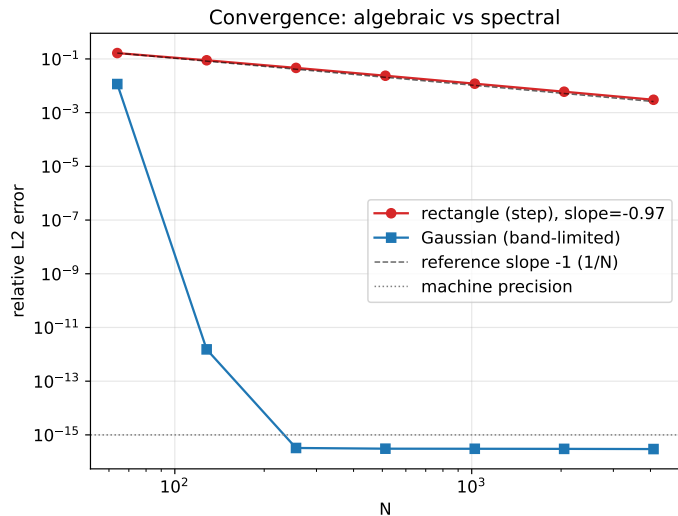
All relative errors are below 1.5%. The circular aperture recovers the first Airy null with relative error 1.74×10^{-4} .

Experiment 1: Two-Dimensional Apertures



The rectangular aperture validates the product sinc pattern. The circular aperture validates the Airy pattern and provides the basis for the Rayleigh recovery experiment.

Experiment 2: Convergence Rates

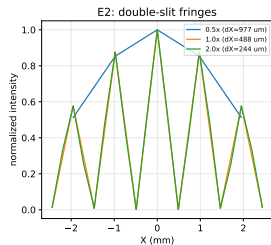
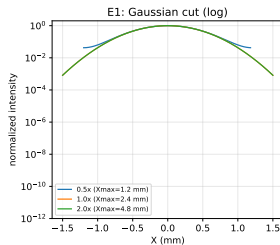
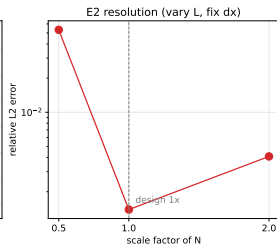
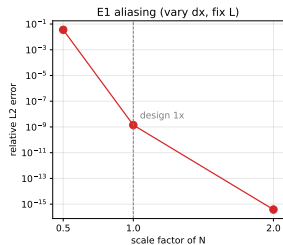


The physical window is fixed and N is increased.

- Square aperture: fitted slope -0.967 , close to $O(1/N)$.
- Gaussian aperture: reaches machine precision by $N = 256$.

Smoothness of the aperture controls spectral decay and convergence.

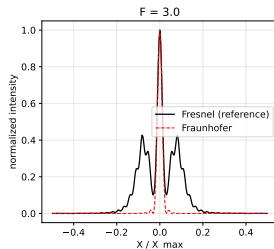
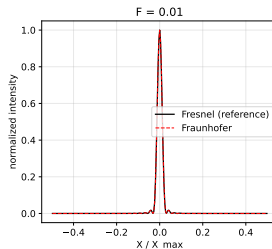
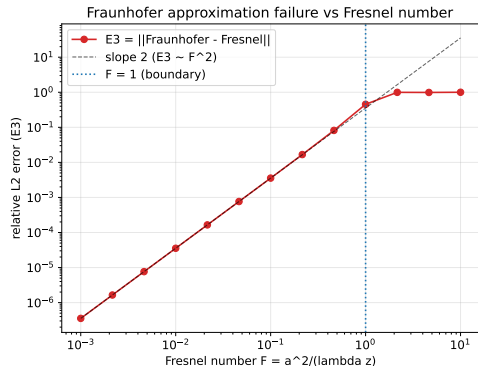
Experiment 3: Grid Design Tightness



Scale	0.5×	1× design	2×
E1 aliasing	3.55%	1.40×10^{-9}	3.79×10^{-16}
E2 resolution	5.33%	0.14%	0.41%

Half-size grids fail visibly; the designed grid is already near the useful threshold.

Experiment 4: Fraunhofer Breakdown



$E3$ is isolated by comparing Fraunhofer with the Fresnel/ASM reference. For small F , the fitted slope is 2.00; around $F = 1$, $E3$ is already about 0.45; for large F , the Fraunhofer pattern is no longer reliable.

Experiment 5: Efficiency

FFT and Simpson quadrature agree to roundoff on the Gaussian test, so runtime is the main difference.

N	FFT time	Simpson time	Speedup
64	$2.5 \times 10^{-4} \text{ s}$	$1.68 \times 10^{-1} \text{ s}$	$\sim 670\times$
128	$9.6 \times 10^{-4} \text{ s}$	2.09 s	$\sim 2200\times$
256	$9.4 \times 10^{-3} \text{ s}$	61.6 s	$\sim 6500\times$

Takeaway

The FFT gives the same numerical result in the tested regime, but its cost scales like $O(N^2 \log N)$ instead of $O(N^4)$.

Experiment 6: Rayleigh Recovery

For a circular aperture, the first Airy dark ring should satisfy

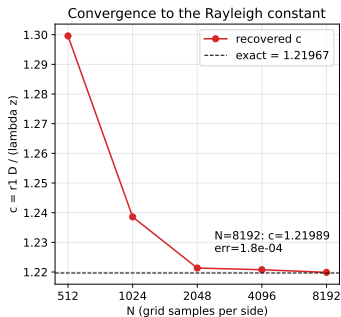
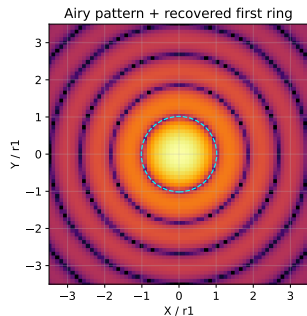
$$r_1 \approx 1.21967 \frac{\lambda z}{D}.$$

The numerical constant is recovered from

$$c_{\text{num}} = \frac{r_1 D}{\lambda z}.$$

The experiment locates r_1 from the central intensity cut using a three-point parabolic refinement.

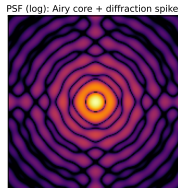
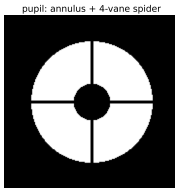
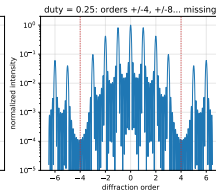
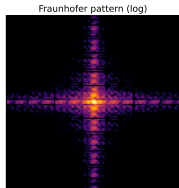
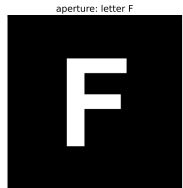
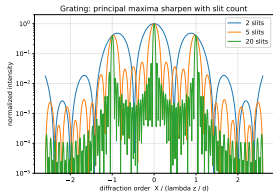
Experiment 6: Numerical Result



N	c_{num}	rel. error
512	1.299592	6.55%
1024	1.238582	1.55%
2048	1.221345	0.137%
4096	1.220777	0.0908%
8192	1.219887	0.0178%

The value converges to 1.21967..., confirming the Rayleigh scale $1.22\lambda/D$.

Experiment 7: Complex Apertures



The same FFT pipeline handles gratings, letter-shaped apertures, and telescope pupils without changing the algorithm.

Main Takeaways

- ① Fraunhofer diffraction reduces the far-field pattern to a Fourier transform of the aperture.
- ② The FFT is efficient, but its grid fixes both screen spacing and observable range.
- ③ E1, E2, and E3 correspond to different mechanisms and must be controlled separately.
- ④ Experiments confirm analytical patterns, convergence behavior, grid tightness, and model breakdown.
- ⑤ The Rayleigh constant $1.21967\dots$ is numerically recovered from the Airy disk.

References



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<https://kmdouglass.github.io/posts/approximating-diffraction-patterns-of-rectangular-apertures-with-the-fft/>.

Thank you

Questions?